

Prepared By	09-Jun-2012	Mr.Chatchawan S.	 Renew coils Engineering & Supply Co., Ltd.	
Checked By	09-Jun-2012	Mr.Chatchawan S.		
Approved By	09-Jun-2012	Mr.Amnauy W.		
Certified By	09-Jun-2012	Mr.Amnauy W.		
Rev.	Date	Checked	Approved	Description
0	09-Jun-2012	Mr.Chatchawan S.	Mr.Amnauy W.	Issue for Approved
1	13-Feb-2013	Mr.Chatchawan S.	Mr.Phuchin S.	Issue for Approved

Renew Coils Engineering & Supply Co., Ltd.

Doc. No. : RNC-WPS-001-2012

Addendum

**WELDING PROCEDURE SPECIFICATION (WPS)
RNC-WPS-005**



WELDING PROCEDURE SPECIFICATION (WPS)

Company Name **Renew Coil Engineering & Supply Co., Ltd.**

WPS No. **RNC-WPS-005**

Date **13-Feb-13**

Revision No. **1**

Supporting PQR No.(s) **RNC-PQR-005**

Date **13-Feb-13**

Type(s) **Manual**

Welding Process(es) **Gas Tungsten Arc Welding (GTAW)**

(Automatic, Manual, Machine, or Semi-Auto.)

JOINT (QW-402)

Joint Design **Single V**

Backing (Yes) ☐ (No) ☒ X

Backing Material (Type) **N/A**

☐ Metal ☐ Nonfusing Metal
☐ Nonmetallic ☐ Other

Notes

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No. **1** Group No. **1** to P-No. **1** Group No. **1**

Material Specification Type and Grade **SA 333 Gr.1.6 SA420 WPL 6** TO **SA 333 Gr.1.6 SA420 WPL 6**

Chem. Analysis and Mech. Prop.

Thickness Range :

Base Metal : Groove **1.6 mm - 12.04 mm.** Fillet **All**

Pipe Dia. Range : Groove **All** Fillet **All**

Other

FILLER METALS (QW-404)

GTAW

Spec. No. (SFA)

5.28

AWS No. (Class)

ER-70S-G

F-No.

6

A-No.

1

Size of Filler Metals

Ø 2.0 & 2.4 mm

Deposited Weld Metals

6.02 mm

Thickness Range :

Groove **12.04 mm Max.**

Fillet **All**

Electrode-Flux (Class)

-

Flux Trade Name

-

Consumable Insert

-

Other

KobeTGS-1N or Equivalents

POSITION (QW-405)

Position(s) of Groove **ALL**

Welding Progression Up ☒ X Down ☐

Position(s) of Fillet **ALL**

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **593 - 649°C**

Time Range **Holding Time 1 Hour**

Notes **Heating & Cooling Rate = 200 ° Max.**

PREHEAT (QW-406)

Preheat Temp. Min. **10°C**

Interpass Temp. Max. **238°C**

Preheat Maintenance **N/A**

Notes

GAS (QW-408)

Percent Composition

	Gas(es)	(Mixture)	Flow Rate
Shielding	Argon	99.99%	13-17Ltr/Min
Trailing	-	-	-
Backing	-	-	-

WPS No. RNC-WPS-005 Rev. 1

ELECTRICAL CHARACTERISTICS (QW-409)

Current ☐ AC ☒ DC Polarity EN
 Amps (Range) 80-120 Volts (Range) 10-13
 Tungsten Electrode Size and Type Ø 2.4 mm, Thoriated (EWTh-2)
 (Pure Tungsten, 2% Thoriated, etc.)
 Mode of Metal Transfer for GMAW N/A
 (Spray arc, short circuiting arc, etc.)
 Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

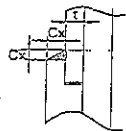
String or Weave Bead BOTH APPLICATIONS
 Orifice or Gas Cup Size 8.0 mm - 19.0 mm
 Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING
 Method of Back Gouging N/A
 Oscillation N/A
 Contact Tube to Work Distance N/A
 Multiple or Single Pass (per side) MULTIPASS
 Multiple or Single Electrodes SINGLE ELECTRODE
 Travel Speed (Range) 5.0-8.0Cm/Minute
 Peening N/A
 Other N/A

Weld Layer(s)	Process	Filler Metal		Current		Volt Range	Travel Speed Range (Cm/min)	Other
		Class	Dia. (mm)	Type Polar.	Amp. Range			
1st	GTAW	ER70S-G	2.4	DC EN	80-100	10-12	5.0-7.0	
2nd & Other(s)	GTAW	ER70S-G	2.4	DC EN	90-120	11-13	6.0-8.0	

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirement of **ASME-IX, 2007 latest Revision.**

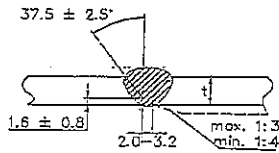
Prepared by: M. Chatchawan S. RNC 13/02/13
 Name Signed Date
 Reviewed by: ภุชชิน สีหอนไชย 22/03/13
 Name Signed Date
 Approved by: ภุชชิน สีหอนไชย 22/03/13
 Name Signed Date

ATTACHED FIGURE JOINT SKETCH



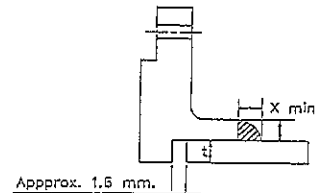
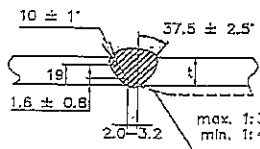
Cx (min.) = $1.25t$ but not less than 3.2 mm.

JOINT No. A1



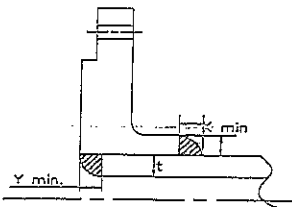
Misalignment (max.) = 1.5 mm.

JOINT No. B1 & B2



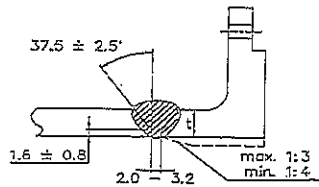
X min. = The lesser of $1.4t$ or thickness of the hub.

JOINT No. A2



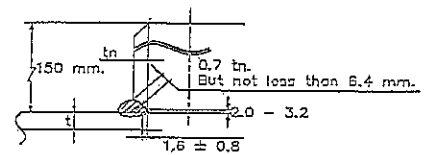
X min. = The lesser of $1.4t$ or thickness of the hub.
Y min. = The lesser of t or 6.4 mm.

JOINT No. A3



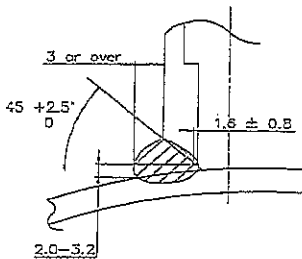
Misalignment (max.) = 1.5 mm.

JOINT No. B3

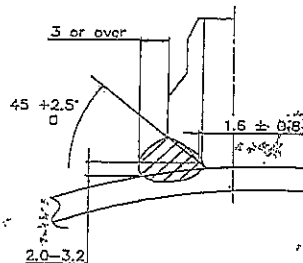


Misalignment (max.) = The lesser of 3.2 mm. or $0.5tn$.

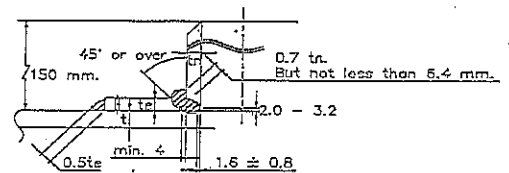
JOINT No. B4



JOINT No. C1

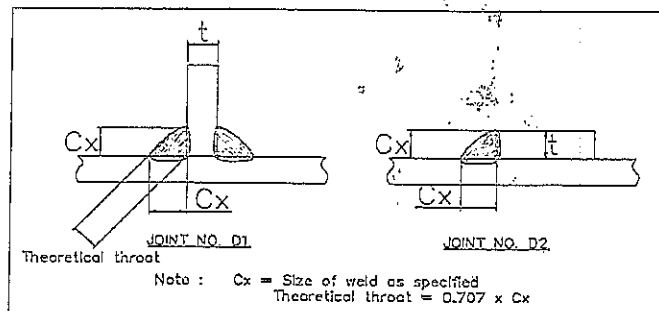


JOINT No. C2



Misalignment (max.) = The lesser of 3.2 mm. or $0.5tn$.

JOINT No. B5



Procedure Qualification Record (PQR)

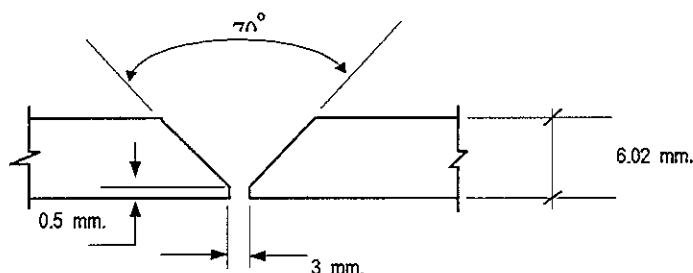
ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

RECORD OF WELDING (QW-483) PQR Number : RNC-PQR-005

Company Name :	RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.		
PQR Record Number :	RNC-PQR-005	Date :	13 February 2013
WPS Number :	RNC-WPS-005		
Welding Process (es) :	Gas Tungsten Arc Welding (GTAW)		
Type (Manual, Semi-auto, or Automatic) :	Manual		

JOINT DESIGN (QW-402)



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

WELDING VARIABLES

BASE METALS (QW-403)				POSTWELD HEAT TREATMENT (QW-407)			
Material Spec.	A 333	To	A 333	Temperature	600° C		
Type or Grade	Gr.6	To	Gr.6	Hold Time	1 Hour		
P. No.	1	To	1	Others	Cooling & Heating Rate 120° C - 195° C (MAX)		
Thickness of Test Coupon		6.02 mm.		GAS (QW-408)			
Diameter of Test Coupon		Ø 4"					
Others							
FILLER METALS (QW-404)				Percent Composition			
SFA Specification	A 5.28				Gas (es)	Mixture	Flow Rate
AWS Classification	ER70S-G			Shielding	Argon	99.99%	15 L/Min.
Filler Metal F-No.	6			Trailing	-	-	-
Weld Metal Analysis A-No.	1			Backing	-	-	-
Size of Filler Metal	Ø 2.4 mm.			ELECTRICAL CHARACTERISTICS (QW-409)			
Others	Kobelco TGS-1N			Current	DC		
Weld Metal Thickness	-			Polarity	EN		
POSITION (QW-405)				Amperage	104 - 132	Voltage	11 - 12
Position of Groove	6G			Tungsten Electrode Size	Ø 2.4 mm. 2% thoriated.EWTh-2		
Progression of Welding (Uphill or Down)	Up Hill			Others	N/A		
Others	N/A			TECHNIQUE (QW-410)			
PREHEAT (QW-406)				Travel Speed	4.26 - 5.28 cm/min.		
Preheat Temperature	26° C			String or Weave Bead	Both		
Interpass Temperature (max.)	189° C			Oscillation	N/A		
Others	N/A			Multiple or single Pass	Multiple Pass		
				Multiple or Single Electrodes	Single Electrode		
				Others	N/A		

ASME-QW-200.2, Section IX (WPQR)

Tensile Test Report No.TS-13-02-086

Guided Bend Tests (QW-160) Report No.BD-13-02-035

Toughness Tests (QW-170)

Filet Weld Test (QW-180)

Type of Test Radiographic examination results Accepted Report No.WQT-005,010

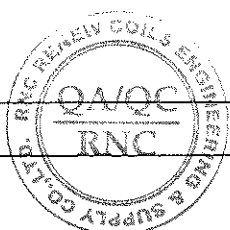
Hardness Testing Report No. HV-13-02-017

Macrostructure Report No. MA-13-02-051

Impact Testing Report No.IM-13-03-030 (-60°C)

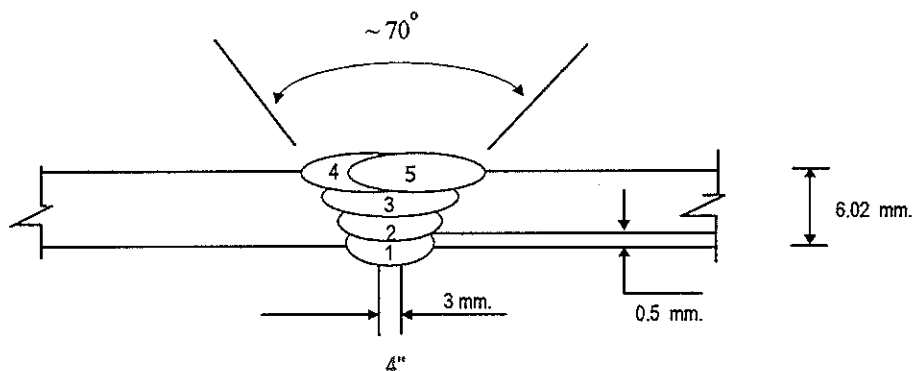
We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

Date : 19 March 2013

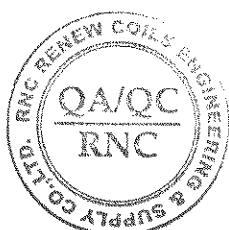


PAE TECHNICAL SERVICE	WELDING PROCEDURE	PQR No. : RNC-PQR-005
		WPS No. : RNC-WPS-005
	QUALIFICATION RECORDS	THIS PQR RECORDS ACTUAL TEST CONDITION AND RESULT

Welding Sequences for PQR-RNC-005(Butt Joint)



Weld Bead No.	Process	Class Filter Metal	Wire Size (mm)	Arc Current (Amps)	Arc Voltage (Volts)	Pot. AC/DC+	String or Weave	Wire Feed	Rate of Travel (cm/min)	Heat Input (KJ/cm)	Gas Shielding Flow Rate (l/min)	Gas Backing Flow Rate (l/min)
							Bead	Speed				
1	GTAW	ER70S-G	2.4	107	11	DCEN	-	-	5.01	14.10	15	-
2	GTAW	ER70S-G	2.4	104	11	DCEN	-	-	5.19	13.23	15	-
3	GTAW	ER70S-G	2.4	132	12	DCEN	-	-	5.28	18.00	15	-
4	GTAW	ER70S-G	2.4	130	12	DCEN	-	-	4.26	21.98	15	-
5	GTAW	ER70S-G	2.4	130	12	DCEN	-	-	4.47	20.94	15	-





PAE TECHNICAL SERVICE CO.,LTD
NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT
 69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250
 Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 607-402-3

WELDER PROCEDURE QUALIFICATION RECORD

Page No. 1 of 1

Client : Renew Coils Engineering & Supply Co.,Ltd.		Project Name : PQR		Location : RNC Work Shop		Report No. PQR-014	
		Customer Order No. : N/A		WPS No. : RNC-WPS-005		Date : 13-Feb-13	

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
W-006	MR.PANJA S.	GTAW	6G	ASME IX	A333Gr.6	ø 4" Thk. 6.02 mm.	A5.28	ACCEPTED	ACCEPTED / PWHT

Joint Details

Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
1	GTAW	ER70S-G	2.4	DCEN	107	11	5.01	26	14.10	Argon	15	-	-
2	GTAW	ER70S-G	2.4	DCEN	104	11	5.19	131	13.23	Argon	15	-	-
3	GTAW	ER70S-G	2.4	DCEP	132	12	5.28	133	18.00	Argon	15	-	-
4	GTAW	ER70S-G	2.4	DCEP	130	12	4.26	150	21.98	Argon	15	-	-
5	GTAW	ER70S-G	2.4	DCEP	130	12	4.47	159	20.94	Argon	15	-	-

PAE Inspection By

Witness By

Approved By

SIGNATURE :

NAME : Mr.Wichit K.

DATE : 14-Feb-2013

SIGNATURE :

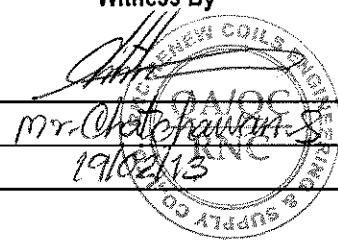
NAME :

DATE :

SIGNATURE :

NAME :

DATE :



PAE TECHNICAL SERVICE CO., LTD.

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

TEL : 02) 322-0222 , FAX : 02) 721-2577 , RAYONG Tel : 038)607402-3, Fax : 038)607402, SRIRACHA : 038)330220, 330229

Page : 1 Of 1[illegible]



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415 , E-mail : qa@lpnmp.co.th

29782

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. TS-13-02-086

Revision No. 0

Test Product	WPS No. RNC-WPS-005	Material Characterization	Pipe
Project Name	WELDER SUPPLY AND PQR	Test Method	ASTM A370-10 (Prepared by ASME IX : 2010)
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A333 Gr.6	Type of Test	Reduced Section Tension
Dimension	Ø 4" (6.02 mm Thk.)	Equipment's Capacity	100 TONS
Date Received	19 February 2013	Equipment / Serial No.	INSTRON / 1000DXR1414
Date Tested	20 February 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm ²)	Gauge Length (mm)	Load (kN)	Strength (N/mm ²)	Load (kN)	Strength (N/mm ²)		
1	547/13 TR1	N/A	5.95	19.14	113.88	N/A	N/A	N/A	51.77	454.62	N/A	OUT OF WELD
2	547/13 TR2	N/A	5.80	19.14	111.01	N/A	N/A	N/A	51.46	463.52	N/A	OUT OF WELD

Additional details : Welding Process / Position : GTAW / 6G

Welder Name : Mr. Panja S.

ID No. 3 2105 00456 92 8

Welder No. : W-006

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 22 / 02 / 13

Approved by :

(Mr.Pavaret Freedawiphat)

Deputy Managing Director

Date 22 / 02 / 13

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29788

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-13-02-035

Revision No. 0

Test Product	WPS No. RNC-WPS-005	Material Characterization	Pipe
Project Name	WELDER SUPPLY AND PQR	Test Method	ASME Sec. IX : 2010 edition
Project No.	N/A	Standart Of Test	ASME Sec. IX : 2010 edition
Material Specification	A333 Gr.6	Type of Jig	Guided-Bend Roller Jig
Dimension	Ø 4" (6.02 mm Thk.)	Diameter of Meudrel	22 mm
Date Received	19 February 2013	Bend Angle	180 Degree.
Date Tested	20 February 2013	Equipment / Serial No.	SHIMADZU UH200A/32098926

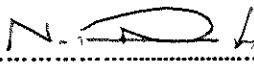
Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	547/13 BF1	Transverse face bend	5.54x18.91x161.32 mm	No Open Discontinuity	N/A	-
2	547/13 BF2	Transverse face bend	5.58x18.90x160.41 mm	No Open Discontinuity	N/A	-
3	547/13 BR1	Transverse root bend	5.56x18.94x160.15 mm	No Open Discontinuity	N/A	-
4	547/13 BR2	Transverse root bend	5.59x19.03x160.32 mm	No Open Discontinuity	N/A	-

Additional details : Welding Process / Position : GTAW / 6G

Welder Name : Mr. Panja S.

ID No. 3 2105 00456 92 8

Welder No. : W-006

Reported by : 

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 22 / 02 / 13

Approved by : 

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 22 / 02 / 13

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13234

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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpm.co.th

Hardness Testing Report

Page 1 of 1

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

Report No. HV-13-02-017

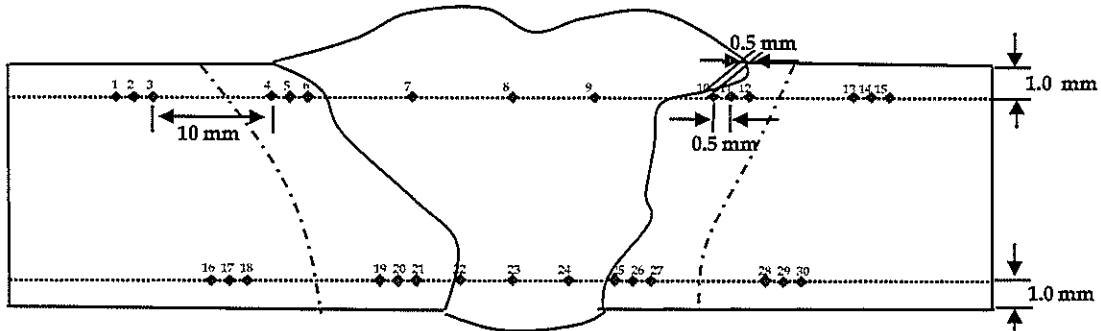
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Revision No. 0

Test Product	WPS No. RNC-WPS-005	Material Characterization	Pipe
Test No.	547/13 HN	Test Method	ASTM E92-82 (Reapproved 2003) ⁶²
Project Name	WELDER SUPPLY AND PQR	Standard of	ASME Section IX : 2010 edition
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	A333 Gr.6	Equipment	Wilson Wolpert , 432 SVD (V32D219)
Dimension	4" (6.02 mm Thk.)	Indenter Size	Pyramid diamond, angle of 136°
Welding Process/Position	GTAW/6G	Test Load	HV 10 Kgf.
Type of Joint	Butt Jiont	Total Force Dwell Time	15 s
Date Received	18 February 2013	Cal. Block	No. A60023 No. A62072
Date Tested	20 February 2013	Ambient Temperature	26.0°C

Test Location

Total = 30 points



Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale)
1	BASE	131.1	16	BASE	132.9	*****		
2	BASE	130.0	17	BASE	131.1			
3	BASE	129.0	18	BASE	132.8			
4	HAZ	136.3	19	HAZ	144.1			
5	HAZ	140.8	20	HAZ	148.8			
6	HAZ	159.3	21	HAZ	151.1			
7	WELD	158.4	22	WELD	144.3			
8	WELD	141.5	23	WELD	142.1			
9	WELD	157.1	24	WELD	142.2			
10	HAZ	158.0	25	HAZ	149.8			
11	HAZ	145.3	26	HAZ	147.5			
12	HAZ	139.7	27	HAZ	143.4			
13	BASE	132.9	28	BASE	130.1			
14	BASE	130.5	29	BASE	132.4			
15	BASE	130.7	30	BASE	132.8			

Additional details :-

Reported by :
(Mr.Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurance
Date : 22/2/2013



Approved by :
(Mr.Pavaret Freedawiphat)
Deputy Managing Director
Date : 22/2/2013

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13698

Macrostructure Analysis Report

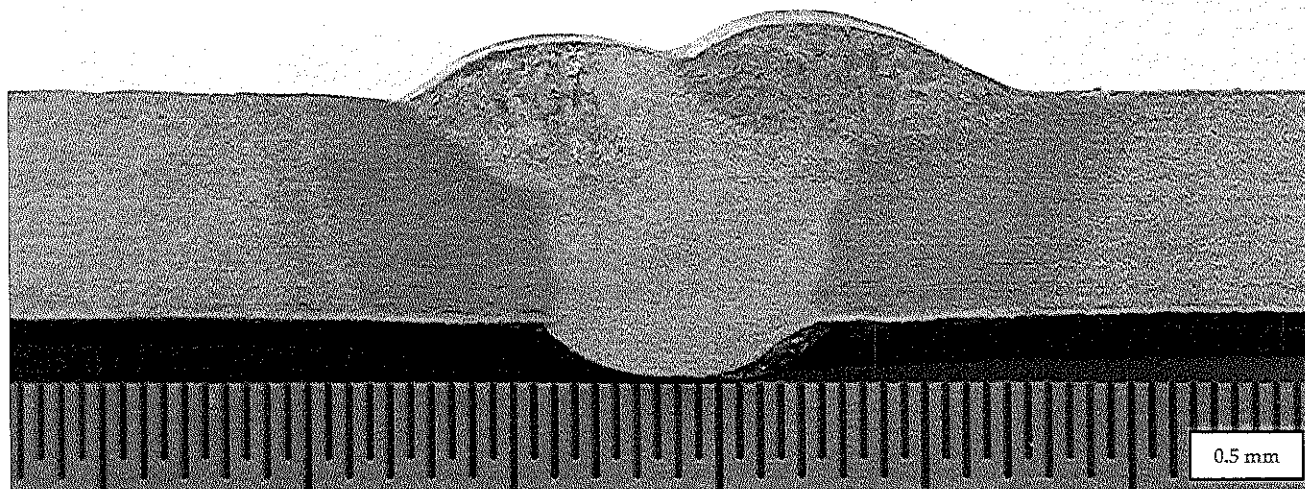
Page 1 of 1

Report No. MA-13-02-051

Revision No.0

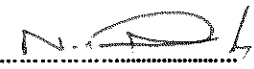
Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.**Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000 Thailand.**

Test Product	WPS No. RNC-WPS-005	Material Characterization	Pipe
Test No.	547/13 MA	Equipment used	Microscope upto 40X (times)
Test Location	-	Serial No.	LEICA 0650854912
Material Specification	A333 Gr.6	Method of Preparation	ASME Secction IX : 2010 edition
Dimension	4" (6.02 mm Thk.)	Test Method	ASME Secction IX : 2010 edition
Welding Process/Position	GTAW/6G	Standard of Test	ASME Secction IX : 2010 edition
Type of Joint	Butt Jiont	Etchant Reagent	25 ml HNO ₃ + 75 ml H ₂ O
Welder Name / No.	Mr. Panja S. / W-006	Time of Etching	5 minutes
Date Received	18 February 2013	Date Tested	20 February 2013



Macro-etch Examination : The macrostructure shows complete fusion of the weldment. Weld metal and heat affected zone are free of cracks.

Additional details : -

Reported by : 
 (Mr.Keerati Sa-ngoen)
 (Acting) General Manager - Quality Assurance
 Date : 22/02/13



Approved by : 
 (Mr.Pavaret Preedawiphat)
 Deputy Managing Director
 Date : 22/02/13

Notes

1. The above results are valid exclusively for tested samples as mentioned in this report.
2. Partial publicity of the results on testing is prohibited without the written permission from LPN LABORATORY.
3. This report shall not be reproduced in whole or in part without the written approval of LPN LABORATORY.

LPN PLATE MILL



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpm.co.th

28667

Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-13-03-030

Revision No. 0

Test Product	WPS No. RNC-WPS-005	Material Characterization	Pipe
Project Name	WELDER SUPPLY AND PQR	Test Method	ASTM A370-10 and ASTM E23M-07a ^{e1}
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A333 Gr.6	Type of Test	Charpy V-Notch Impact
Dimension	Ø 4" (6.02 mm Thk.)	Equipment / Serial No.	Tinius Olsen / 221811
Received Date	18 February 2013	Equipment 's Capacity	542 Joules
Tested Date	18 March 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Specimen Dimension			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	547/13-1	-60°C	3.310	10.019	54.77	17.79	19.39	Base Metal	-
2	547/13-2	-60°C	3.309	10.018	54.75	22.11			
3	547/13-3	-60°C	3.312	10.019	54.78	18.28			
4	547/13-1	-60°C	3.310	10.018	54.90	35.24	33.71	HAZ	-
5	547/13-2	-60°C	3.313	10.018	54.88	35.34			
6	547/13-3	-60°C	3.311	10.017	54.88	30.54			
7	547/13-1	-60°C	3.306	10.018	54.76	56.91	60.44	Weld Metal	-
8	547/13-2	-60°C	3.310	10.016	54.70	58.54			
9	547/13-3	-60°C	3.316	10.017	54.79	65.88			

Additional details : Welding Process / Position : GTAW / 6G

Welder Name : Mr. Panja S.

ID No. 3 2105 00456 92 8

Welder No. : W-006

Reported by :

(Mr. Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date/...../.....

Approved by :

(Mr. Pavaret Preedawiphat)

Deputy Managing Director

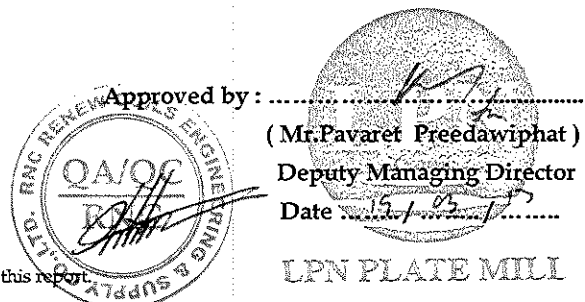
Date/...../.....

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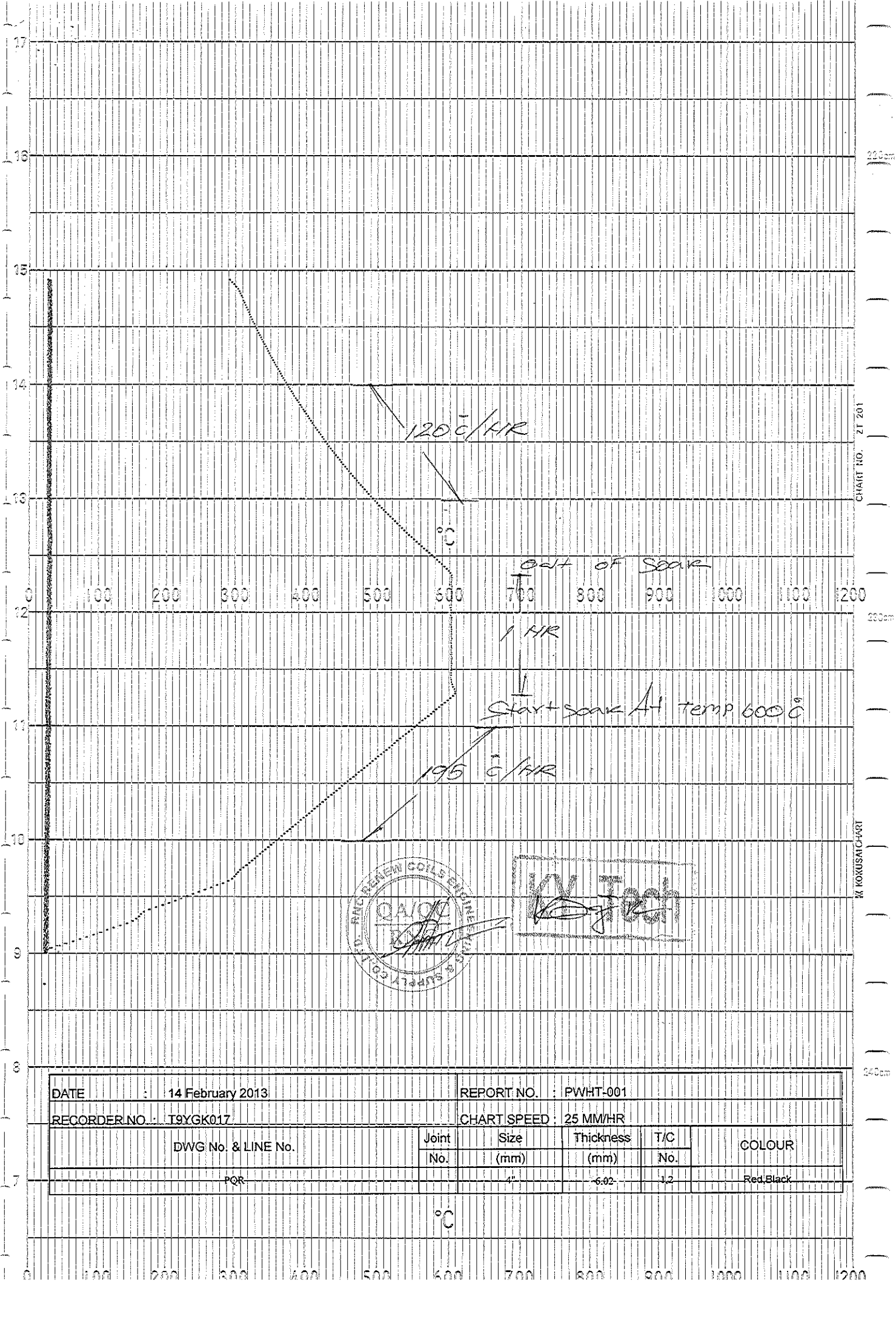
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4. This report is tested for LPN Metallurgical Research Center (Thailand) Co., Ltd.



LPN PLATE MILL

	CHECKED BY	REVIEWED BY	APPROVED BY
COMPANY	R & Tech Co., Ltd	RENEWCOILS ENGINEERING & SUPPLY CO., LTD.	
SIGN			
NAME	MR. PANYOM KONGPOO	Mr. Chatchawan S.	ภูชิน สีหอนไชย
DATE	15 February 2013	12/05/13	Phuchin Seehomchai



DATE : 14 February 2013		REPORT NO. : PWHT-001			
RECORDER NO. : T9YGK017		CHART SPEED : 25 MM/HR			
DWG No. & LINE No.	Joint	Size	Thickness	T/C	COLOUR
	No.	(mm)	(mm)	No.	
PQR		4"	6.02	1.2	Red/Black



K Y TECH CO., LTD

1/132 Moo 1 T. Phra, A. Banchang Rayong 21130

Tel: 0066 38 025158

Fax: 0066 38 025186

Certificate of Calibration

Instrument: Temperature Recorder
Model: EH100-12
Serial No.: T9YGK017
Manufacturer: CHINO
Measuring Range: 0~1200 °C
Calibration Date: 1 February 2013
Due Date: 31 July 2013

Cert. No. : KY-023-2013

KY

Condition of this result of test

Reference Standard Instrument

Instrument	Model	Serial No.	Certificate Of. Calibration No.	Calibration Date	Due Date
DAILCHI	TC-520	-	TC07257	29 February 13	28 February 14

Tech

Result Of Calibration

Input °C	100	200	300	400	500	600	700	800	900	1000
Reading °C	100	200	301	401	500	601	701	802	900	1000

Calibrated By

KY Tech
(Mr. Tawat Chongprong)



K Y TECH CO., LTD

1/132 Moo 1 T. Phra, A. Banchang Rayong 21130

Tel: 0066 38 882715

Fax: 0066 38 882715

Certificate of Calibration

Instrument: Temperature Recorder
Model: EH100-12
Serial No.: T9YGK016
Manufacturer: CHINO
Measuring Range: 0~1200 °C
Calibration Date: 1 February 2013
Due Date: 30 July 2013

Cert. No. : KY-030-2013

KY

Condition of this result of test

Reference Standard Instrument

Instrument	Model	Serial No.	Certificate Of. Calibration No.	Calibration Date	Due Date
DAILCHI	TC-520	-	TC07257	29 February 13	28 February 14

Result Of Calibration

Recorder Input type "K"

Input °C	100	200	300	400	500	600	700	800	900	1000
Reading °C	100	200	301	401	500	601	701	802	900	1000

Calibrated By


(Mr. Tawan Teupong)

CERTIFICATE OF CALIBRATION



Issued By Marske Machine Hire & Calibration Centre

Date of Issue January 29, 2013

Certificate Number
T30065

Page 1 of 2 Pages

Marske Machine (Thailand) Co., Ltd
2/58 Highway 3191, MapTaPhut, Muang, Rayong 21150
038 682 992 - 3, 038 691 563 - 4
Fax : 038 682 988, 038 681 173
email : sales@mmc-thai.com

Approved Signatory

☒ Marut L. / TM

Customer : K Y Tech Co., Ltd.
1/132 Moo 1 T. Phra, A. Banchang Rayong 21130

Date Received : January 28, 2013

Instrument -	System ID :	TC07257	Job Number :	0000130570
	Description :	Thermometer & Calibrator	Ref. Number :	-
	Manufacturer :	DAIICHI		
	Model Number :	TC-520		
	Serial Number :	-	Last Certificate Number :	T20184
	Procedure Version :	S2/1	Last Calibration Date :	08/02/2012

Environmental Conditions

Temperature : 20°C +/- 2°C
Relative Humidity : 55% +/- 10%

Mains Voltage : 220V +/- 10V
Mains Frequency : 50Hz +/- 1Hz

Comments

- Instrument was placed in lab and allowed to stabilise before calibration.
- The Instrument has been calibrated by signal simulation technique.
- The instrument tolerance is +/- 3°C
- All test passed specification.

Traceability Information

Instrument description	Serial number	Certificate number	Cal. Date	Cal. Period
Precision Calibrator M2001#1	M2001#1	062LTE-6	07/12/2012	52

Calibrated By : Sarayut Sarayut

Date of Calibration : January 29, 2013

This certificate provides traceability of measurement to recognised National Standards, and to the units of measurement realised at the National Physical Laboratory or other recognised National Standards laboratories.

Copyright of this certificate is owned by the issuing laboratory and may not be reproduced except with the prior written approval of the issuing laboratory. This certificate complies with the requirements of BS EN ISO 10012:2003.

CERTIFICATE OF CALIBRATION

mmc[®]
(THAILAND) LTD

Certificate Number
T30065

Page 2 of 2 Pages

Test Title	Tolerance	STD Value	UUC Reading	% Of Spec.	Uncertainties
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Calibration Information.

The uncertainty limits quoted refer to the measured values only, with no account being taken of the instrument ability to maintain its calibration.

The reported expanded uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by a coverage factor $k=2$ corresponding to a coverage probability of approximately 95%.

Function :Measurement Mode

T/P -100°C	3°C	-100.0°C	-99°C	33%	0.6°C
T/P -50°C	3°C	-50.0°C	-49°C	33%	0.6°C
T/P 0°C	3°C	0.0°C	0°C	0%	0.6°C
T/P 50°C	3°C	50.0°C	50°C	0%	0.6°C
T/P 100°C	3°C	100.0°C	101°C	33%	0.6°C
T/P 500°C	3°C	500.0°C	499°C	33%	0.6°C
T/P 1000°C	3°C	1 000.0°C	999°C	33%	0.6°C
T/P 1200°C	3°C	1 200.0°C	1 199°C	33%	0.6°C

Function :Calibration Mode

T/P -100°C	3°C	-100.36°C	-100°C	12%	0.6°C
T/P -50°C	3°C	-50.95°C	-50°C	32%	0.6°C
T/P 0°C	3°C	0.51°C	0°C	17%	0.6°C
T/P 50°C	3°C	49.43°C	50°C	19%	0.6°C
T/P 100°C	3°C	99.07°C	100°C	31%	0.6°C
T/P 500°C	3°C	500.63°C	500°C	21%	0.6°C
T/P 1000°C	3°C	1 001.26°C	1 000°C	42%	0.6°C
T/P 1200°C	3°C	1 201.63°C	1 200°C	54%	0.6°C

End of Calibration